

OSSP SKILL 3

K.Sai Akhil

ROLL NUMBER : 2520030021

3) To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#define MAX 10
int main()
{
    char *history[MAX];
    char command[100];
    int count = 0;
    while (1)
    {
        printf("shell> ");
        fgets(command, sizeof(command), stdin);
        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
            break;

        if (strcmp(command, "history") == 0)
        {
            for (int i = 0; i < count; i++)
                printf("%d: %s\n", i + 1, history[i]);

            continue;
        }
        if (count < MAX)
        {
            history[count] = malloc(strlen(command) + 1);
            strcpy(history[count], command);
            count++;
        }
    }
    for (int i = 0; i < count; i++)
        free(history[i]);
    return 0;
}
```

Output :

```
akhil@Akhil:~/skill-1$ gcc skill3a.c -o skill3a
akhil@Akhil:~/skill-1$ ./skill3a
akhil$> hi
akhil$> OS COURSE
akhil$> KLU
akhil$> history
1: hi
2: OS COURSE
3: KLU
akhil$> exit
akhil@Akhil:~/skill-1$ |
```

4) To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

```
#include <stdio.h>
#include <string.h>

int main()
{
    char input[100];
    char *token;

    printf("Enter command: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    if (strlen(input) == 0)
    {
        printf("Empty command\n");
        return 0;
    }

    token = strtok(input, " ");

    while (token != NULL)
    {
        printf("Token: %s\n", token);
        token = strtok(NULL, " ");
    }

    return 0;
}
```

OUTPUT:

```
akhil@Akhil:~/skill-1$ ./skill3b
Enter command: ls 1
Token: ls
Token: 1
akhil@Akhil:~/skill-1$ |
```